

Statistics

Statistical Thinking

Statistics is the science of learning from data.

Statistics is a set of equations that allows us to solve complex problems.

But very simple and practical definition of statistics is: -.

"Statistics is the science of making decisions under uncertainty."

How statistics can be used in everyday life?

Uses of statistics in day to day life

Statistical Thinking

EXAMPLE 1:-

“Vacation in Goa , you are Renting a basic Normal Bike for one day local sight-seeing , do you want to buy the bike insurance for Rs. 500/per day ?”

Example 2:-

My friend wise grandfather smoked 5 packs a day and drank a quart of scotch a day, but was always healthy and died peacefully in his sleep when he was 90,

Do you conclude that all the health warnings about cigarettes are wrong?

Example 3:-

A celebrity advertised Amul ice-cream in Month May last year, due to that ice-cream sales increase 25% in following three months. Thus the advertisement was effective.

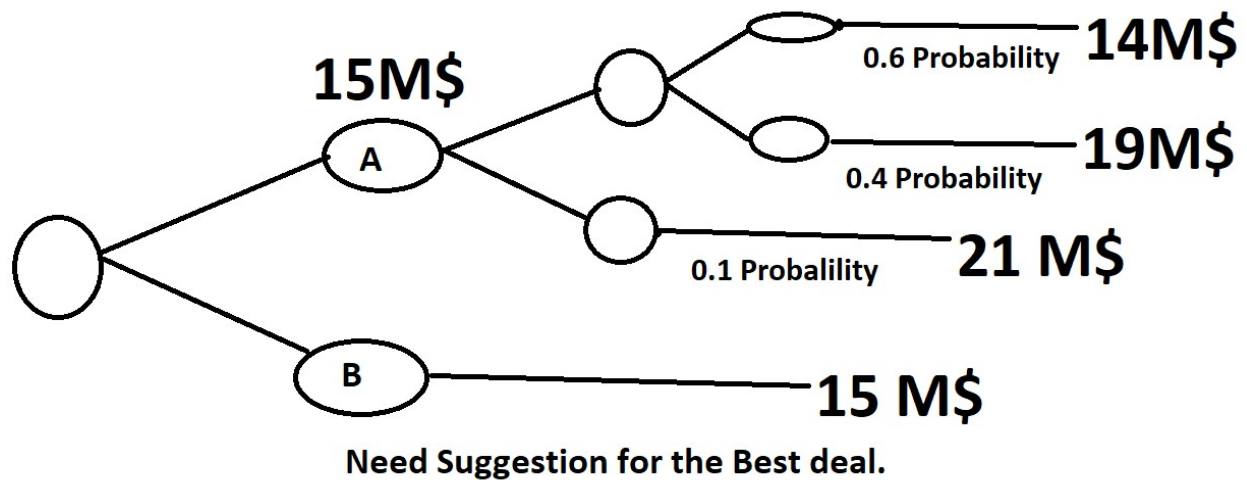
Example 4:-

The more Liquor shop in the city , the more crime there is , so the liquor shop is the responsible for crime?

Statistics is very useful for quantifying uncertainty which we face in decisions we make every day.

One example is risk aversion.

- On spending \$100 Would you rather win \$110 guaranteed or **50% chance of winning \$140 and 50% chance of losing \$20?**
- On spending \$100 Would you rather win \$107 guaranteed or **70% chance of winning \$130 and 30% chance of losing \$20?**



recent example: -

In this pandemic time of covid-19, statistics is used widely.

- Perdition of Covid-19 Phase 1 , Phase II , Covid-19 New variant and its impact possibilities etc..
- To determine the number of people who got vaccinated and who are left, we use statistics to obtain this data.

Recent state Election News 24-Today's Chanakya. Several exit polls have predicted a massive victory for the political party A in the state. The ruling party is projected to win 140-150 seats out of the total 182 seats in state Assembly by different exit polls.

- What percentage of voters approve the Political Party A handling the Job?
- Is this difficult to determine exactly as there are more than 7 crore population in Gujrat.?

Population & Sample

- ➔ A population is the entire group that you want to draw conclusions about.
- ➔ A sample is the specific group that you will collect data from.

A well-chosen sample will contain most of the information about a particular population parameter.

Measures of Central Tendency

In statistics, there are three common measures of central tendency:

- **The mean**
- **The median**
- **The mode**

Mean = (sum of all values) / (total no. of values)

Example: -home runs hit by 10 baseball players on the same team in one season:

Player	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Home Runs	8	15	22	21	12	9	11	27	14	13

The mean number of home runs hit per player can be calculated as:

Mean = $(8+15+22+21+12+9+11+27+14+13) / 10 = 15.2$ home runs.

The **median** is the middle value in a dataset (arranged in ascending order)

For example, to find the median number of home runs hit by the 10 baseball players

Player	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Home Runs	8	15	22	21	12	9	11	27	14	13

we can arrange the players in ascending order:-

Player	#1	#6	#7	#5	#10	#9	#2	#4	#3	#8
Home Runs	8	9	11	12	13	14	15	21	22	27

Since we have an even number of values, the median is simply the average of the two middle values: **13.5**.

Instead, consider if we had nine players:

Player	#1	#6	#7	#5	#9	#2	#4	#3	#8
Home Runs	8	9	11	12	14	15	21	22	27

In this case, since we have an odd number of values the median is simply the middle value: **14**.

The **mode** is the value that occurs most often in a dataset

The **mode** is the value that occurs most often in a dataset. A dataset can have no mode (if no value repeats), one mode, or multiple modes.

For example, the following dataset has no mode:

Player	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Home Runs	8	9	11	12	13	14	15	21	22	27

The following dataset has one mode: **15**. This is the value that occurs most frequently.

Player	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Home Runs	8	9	11	12	13	15	15	21	22	27

The following dataset has three modes: **8, 15, 19**. These are the values that occur most frequently.

Player	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Home Runs	8	8	11	12	15	15	17	19	19	27

The mode can be a particularly helpful when working with categorical data because it tells us which category occurs most frequently.

A young couple is interested to buy their first home in a new city and their budget is \$150,000. Some societies in the city have expensive houses, some have cheap houses, and others have medium-priced houses. They want to easily narrow down their search to specific society that are within their budget.

If the couple just looked at the individual home prices in each society, they might have a tough time determining which society best fit their budget because they might see something like this:

society A home prices: \$140k, \$190k, \$265k, \$115k, \$270k, \$240k, \$250k, \$180k, \$160k, \$200k, \$240k, \$280k, ...

society B home prices: \$140k, \$290k, \$155k, \$165k, \$280k, \$220k, \$155k, \$185k, \$160k, \$200k, \$190k, \$140k, \$145k, ...

society C home prices: \$140k, \$130k, \$165k, \$115k, \$170k, \$100k, \$150k, \$180k, \$190k, \$120k, \$110k, \$130k, \$120k, ...

When to Use the Mean, Median, and Mode

Mean: Finds the average value in a dataset.

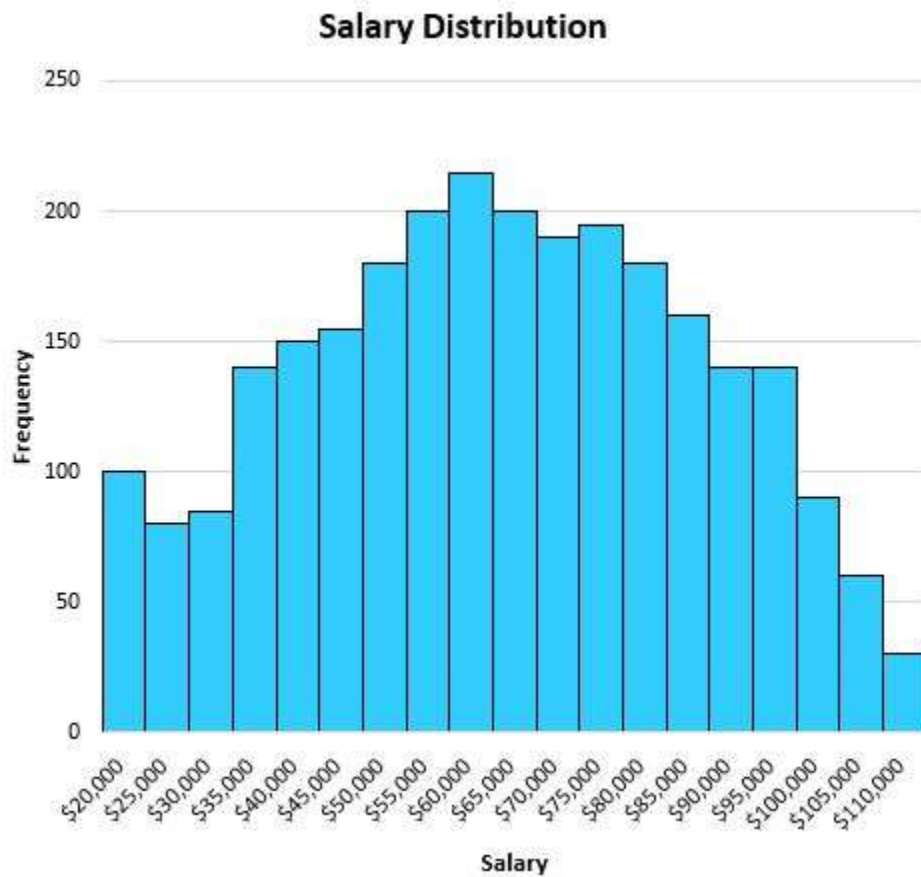
Median: Finds the middle value in a dataset.

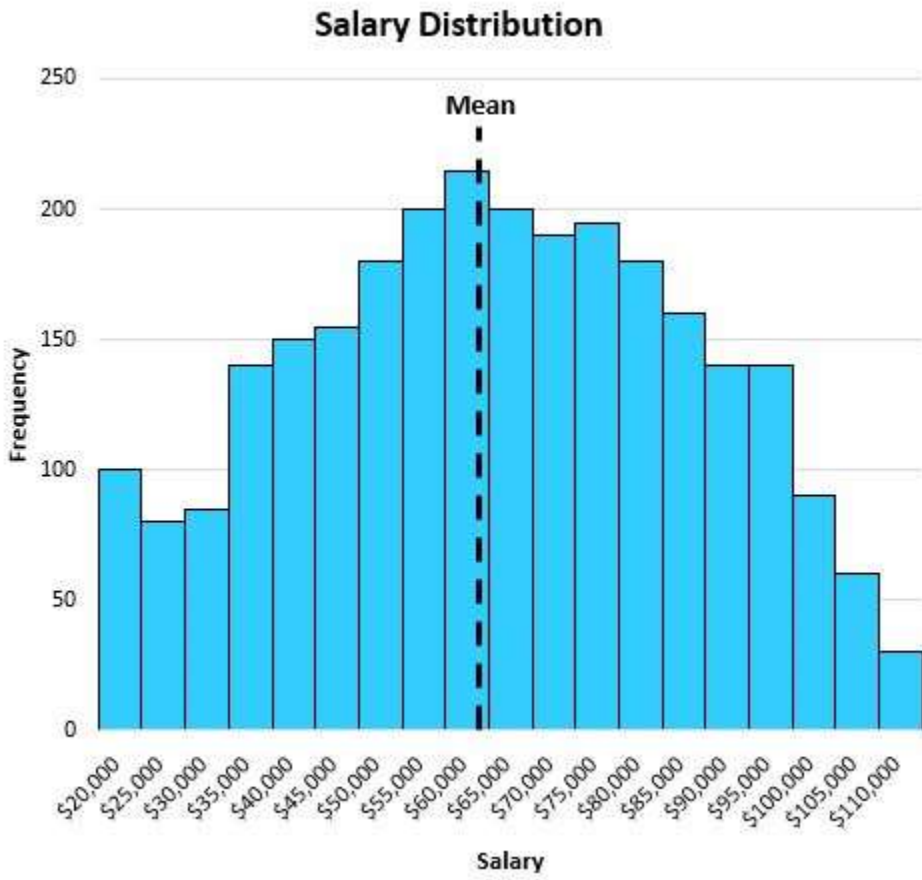
Mode: Finds the most frequently occurring value in a dataset.

When to use the mean

use the mean when the distribution of the data is fairly symmetrical and there are no outliers.

For example, salaries of individuals in a certain town:



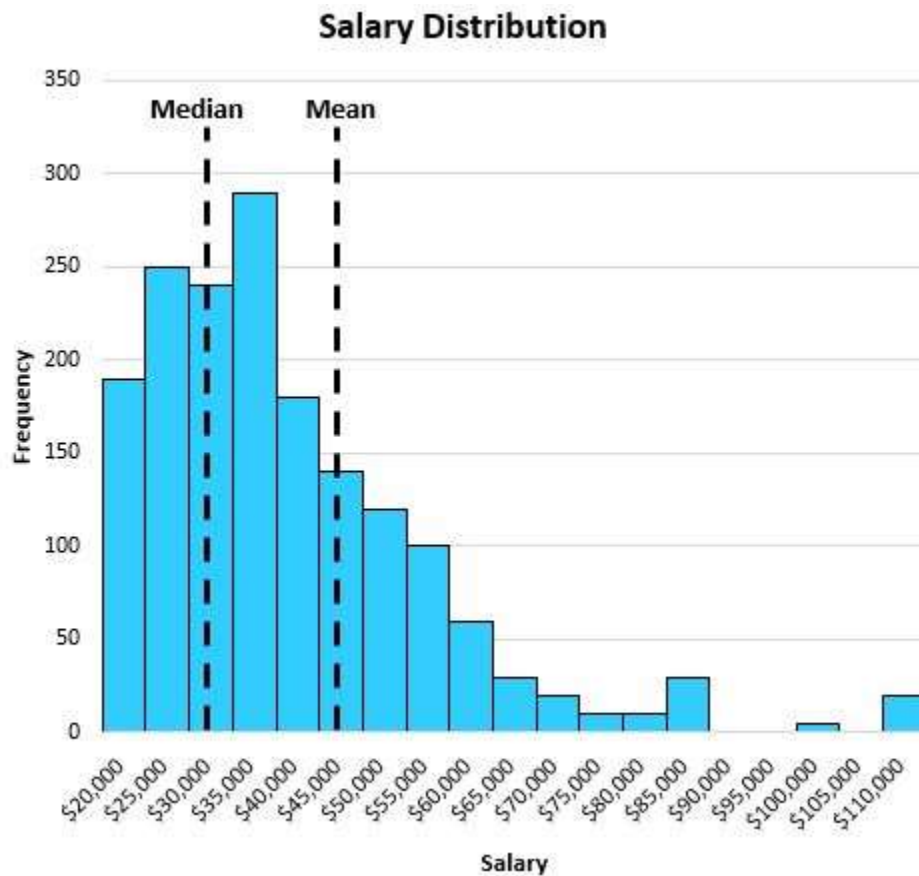


When to use the median

It is best to use the median when the distribution of the data is either skewed or there are outliers present.

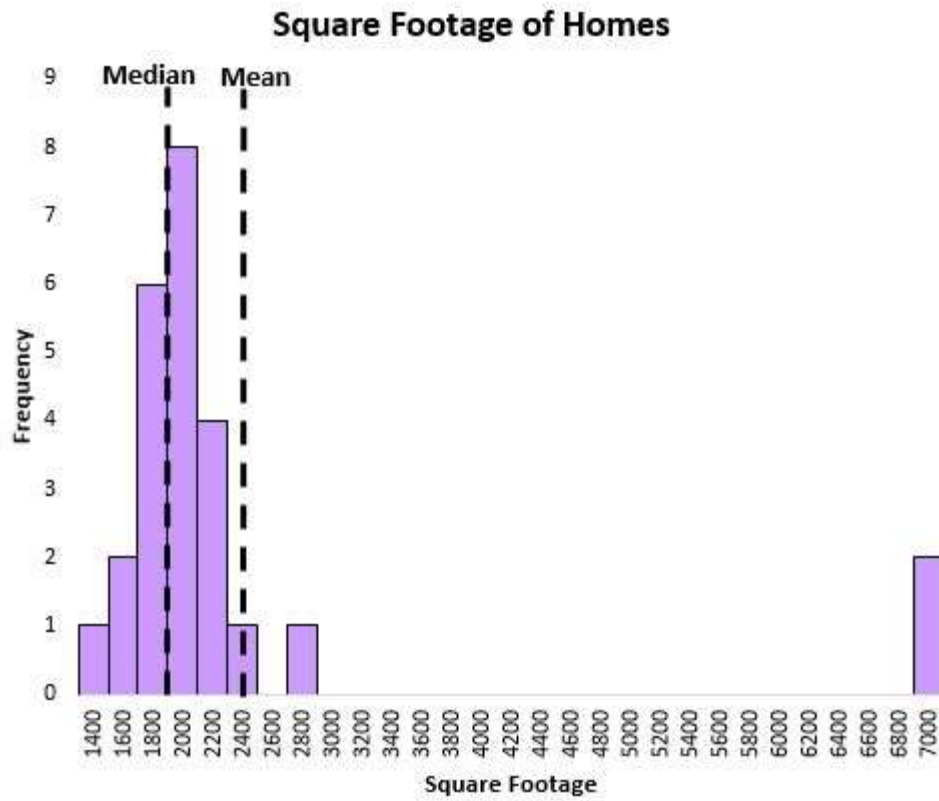
Skewed data:

For example, distribution of salaries for individuals in a certain town:



Outliers:

For example, the square footage of houses on a certain street:



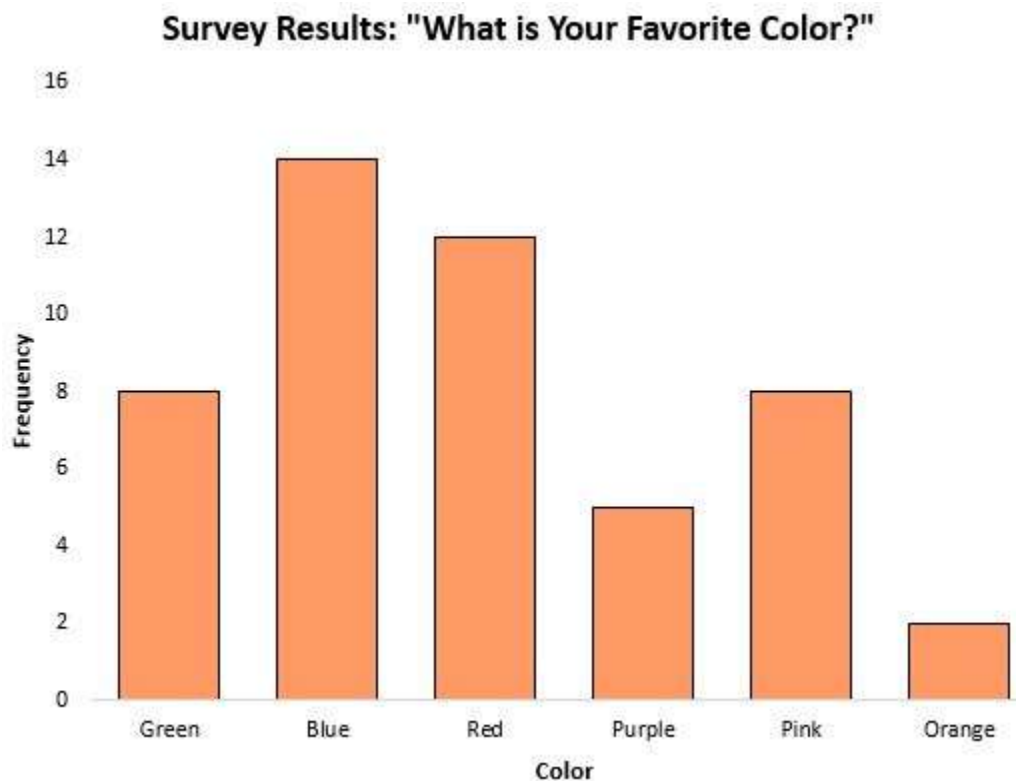
When to use the mode

It is best to use the mode when working with categorical data and want to know which category occurs most frequently.

examples:-

conduct a survey about people's favorite colors and want to know which color occurs most frequently in the responses.

conduct a survey about people's preferences among three choices for a website design and want to know which design people prefer most.



working with categorical data then it's not even possible to calculate the median or mean, which leaves the mode as the only measure of central tendency.